

FORMALISATION AND EVALUATION OF PROPERTIES FOR CONSEQUENTIALIST MACHINE ETHICS

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INTRODUCTION

Motivation

- AI systems increasingly make decisions that impact humans' lives (self-driving cars)
- Need to follow ethical principles

Contexte

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Problem

- Existing approaches lack rigorous justification and formal comparison.

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- consequentialism: judges actions only based on their **consequences**
- deontology: notion of **duty** to determine whether an action/sequence of actions is right or wrong
- virtue ethics: uses agent's **character and intention** to determine the appropriateness of their actions

Ethics

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- virtue ethics: uses agent's **character and intention** to determine the appropriateness of their actions

We will focus on **consequentialism** because it is the dominant approach to machine ethics so far and the focus of this paper.

Question

How can machine ethics approaches properly validate their formalisation of ethical principles and determine the most appropriate principle(s) to apply in a given situation?

Problematique

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Contributions

- Framework to evaluate ethical principles systematically,
- Define formal principles and properties for decision-making.

Running example

*We have available to us exactly **one smart robot**, designed to assist in decision-making when “incidents” occur. A **fire just started** on the top floor of a building **full of employees**. The robot needs to **keep everyone safe** but also to **minimise the damage by extinguishing the fire**. It has three choices:*

- *(a) rush into the source of the fire and extinguish it,*
- *(b) personally help the evacuation of employees until everyone is safe,*
- *(c) trigger the fire alarm while calling the fire department.*

What should it do?

SITUATION CALCULUS

What is it?

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Lets define the framework!

Fluents

What they are: Properties of the world that can change over time,

Example: "The building is on fire", "Employees are safe", "The fire is extinguished",

Formal notation: $F(\vec{x}, s)$ means "property F with arguments $\vec{x}$ is true in situation s ."

Definitions

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Situations

What they are: Snapshots of the world at a given time, including the history of actions that led to it,

Example:

- s_0 : Initial situation (building on fire, employees inside)
- $do(a, s)$: The situation after performing action a in situation s .

Formal notation: $do([evac, extg], s_0)$ means the situation after evacuating and then extinguishing the fire.

Actions

What they are: Things that change the world,

Example: *evac* (evacuate employees), *extg* (extinguish fire), *call* (call the fire department).

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Action Preconditions

What they are: Conditions that must be true for an action to be possible.

Example: To *extg*, the robot must be near the fire source.

Formal notation: $Poss(a(\vec{x}), s) \equiv \pi_a(\vec{x}, s)$ means "action a is possible in situation s if π_a is true."

Effect Axioms

What they are: Rules that describe how actions change fluents.

- Positive effect: F becomes true after a .
- Negative effect: F becomes false after a .

Example:

- $do(evac, s)$ makes "employees are safe" true.
- $do(extg, s)$ makes "building is on fire" false.

Formal notation:

- $\gamma_F^+(\vec{x}, a, s) \rightarrow F(\vec{x}, do(a, s))$ (positive effect)
- $\gamma_F^-(\vec{x}, a, s) \rightarrow \neg F(\vec{x}, do(a, s))$ (negative effect)

Successor State Axiom

What it is: A rule that fully describes how a fluent changes after an action,

Example: If *extg* is performed, "building is on fire" becomes false, and all other fluents stay the same unless changed,

Formal notation:

$$F(\vec{x}, do(a, s)) \equiv \gamma_F^+(\vec{x}, a, s) \vee (F(\vec{x}, s) \wedge \neg \gamma_F^-(\vec{x}, a, s))$$

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Plans

What they are: Sequences of actions,

Example: $\pi = [evac, extg, call]$,

Formal notation: $do(\pi, s_0)$ is the situation after executing plan π from s_0 .

Updated Example

Consider an initial state of our example where the building is on fire and all employees are inside. The robot has options to extinguish the fire (extg), evacuate all employees (evac), and call the fire department (call). The employees will get injured if not evacuated immediately and the building will become more damaged the longer it is not extinguished.

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Let's try and give some examples using the previous definition!

- *OnFire*(*s0*): True (the building is on fire),
- *EmployeesSafe*(*s0*): False (employees are not safe),
- *FireExtinguished*(*s0*): False (the fire is not extinguished),

Examples

In s_0 , we can say that:

- $OnFire(s_0)$: True (the building is on fire),
- $EmployeesSafe(s_0)$: False (employees are not safe),
- $FireExtinguished(s_0)$: False (the fire is not extinguished),

And some action preconditions could be:

- $Poss(extg, s) \equiv OnFire(s)$: The robot can extinguish the fire only if the building is on fire,
- $Poss(evac, s) \equiv \neg EmployeesSafe(s)$: The robot can evacuate employees only if they are not already safe.

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And one positive effect axiom could be:

- $\gamma_{FireExtinguished}^+(extg, s) \rightarrow FireExtinguished(do(extg, s))$
which means that performing the extinguish action in s always results in $FireExtinguished$ to become True.

FORMALISATION OF ETHICAL PRINCIPLES

Definition

Overview

Objective here is not to achieve a **goal**, but rather to see if a plan satisfies an **ethical principle**.

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Goodness

We define a plan as **ethically permissable** based on its *goodness*.
We introduce the function *goodness(state(s))* which returns a *value*.

We also introduce *harmful(φ)* which is true if φ is a harmful fluent.

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We also introduce $harmful(\varphi)$ which is true if φ is a harmful fluent.

We simplify the notation for *admissible* plans and goodness:

- $state(\pi) = state(do(\pi, s_0))$
- $goodness(\pi) = goodness(state(\pi))$

Admissibility

Ethical Admissibility

We define a plan π as **ethically admissible** if it is both possible and satisfies an ethical principle.

The paper defines 5 ethical principles: **utilitarianism**, **hedonism**, **no new harm**, **remove harm** and **harm avoidance**.
Let's define them formally!

Utilitarianism

It focuses on bringing the greatest good to the greatest number of people:

$$\mathcal{EP}_{util}(\Pi, s_0) = \{\pi \in \Pi \mid goodness(\pi) \geq goodness(\pi') \text{ for all } \pi' \in \Pi\}$$

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Example:

If $\text{goodness}([\text{evac}, \text{extg}]) > \text{goodness}([\text{extg}, \text{call}])$, then $[\text{evac}, \text{extg}]$ is ethically permissible.

Note: $\mathcal{EP}$ stands for ethical permissibility function.

$$\mathcal{EP}_{hedon}(\Pi, s_0) = \{\pi \in \Pi \mid goodness(\pi) \geq goodness(\emptyset)\}$$

Hedonism

Based on the overall consequences that it produces, a direct interpretation of consequentialism: any plan is ethically permissible if it has more goodness at the end state compared to the initial state.

$$\mathcal{EP}_{hedon}(\Pi, s_0) = \{\pi \in \Pi \mid goodness(\pi) \geq goodness(\emptyset)\}$$

Example:

If $goodness([evac]) \geq goodness(\emptyset)$, then $[evac]$ is ethically permissible.

No New Harm

It dictates that a plan should not cause any new harm, so an agent should not be directly (or actively) responsible for any harm. This also means that we tolerate a harm that is unavoidable.

$$\mathcal{EP}_{nnHarm}(\Pi, s_0) = \{\pi \in \Pi \mid \text{for all } \varphi, \text{ if } harmful(\varphi) \text{ and } state(s_0) \not\models \varphi, \\ \text{then } (state(\pi) \not\models \varphi \text{ or for all } \pi' \in \Pi, state(\pi') \vdash \varphi)\}$$

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For every harmful predicate φ ,

If φ is not true in the initial state ($state(s_0) \not\models \varphi$), Then φ must not become true after executing π ($state(\pi) \not\models \varphi$), unless φ becomes true after every admissible plan (for all $\pi' \in \Pi$, $state(\pi') \models \varphi$), i.e unavoidable.

Remove Harm

Remove Harm

A plan is ethically permissible if it removes existing harm in the initial situation. It requires the agent to mitigate harm whenever possible.

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Explanation:

For every harmful predicate φ , if it is found in the initial state ($\textit{state}(s_0) \vdash \varphi$), then an acceptable plan π must remove it, or every possible plan must have it at the end (so unavoidable).

Harm Avoidance

Harm Avoidance

A plan is ethically permissible if it minimises the amount of harm in the end state.

$$\mathcal{EP}_{\text{harmAvoid}}(\Pi, s_0) = \{\pi \in \Pi \mid \text{for all } \varphi, \text{ if } \textit{harmful}(\varphi), \\ \text{then } (\textit{state}(\pi) \not\vdash \varphi \text{ or for all } \pi' \in \Pi, \textit{state}(\pi') \vdash \varphi)\}$$

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Explanation:

For every harmful predicate φ , an acceptable plan π must ensure that φ is not true in the end state, or every possible plan must have it at the end (so unavoidable). It is a combination of the two previous principles.

Example

Let's define that the most important thing is to keep the employees safe. So we define the world as:

- $inside(emp, do(a, s)) \equiv inside(emp, s) \wedge a \neq evac$
- $onfire(do(a, s)) \equiv onfire(s) \wedge a \neq extg$
- $injured(emp, do(a, s)) \equiv inside(emp, s) \wedge onfire(s) \wedge a \neq evac$
- $injured'(emp, do(a, s)) \equiv injured(emp, s) \wedge a \neq evac \wedge inside(emp, s) \wedge onfire(s)$

Here *emp* is the employee and *injured*(*emp*) the penalty of letting someone get injured. *injured'*(*emp*) means even more damage to an employee.

Example

According to our previous definitions, we model the goodness function as:

- | | |
|----------------------------------|--|
| ■ $goodness([call]) = -10$ | ■ $goodness([extg, evac]) = -10$ |
| ■ $goodness([extg]) = -10$ | ■ $goodness([call, evac, extg]) = -10$ |
| ■ $goodness([evac]) = 0$ | ■ $goodness([evac, call, extg]) = 0$ |
| ■ $goodness([call, evac]) = -10$ | ■ $goodness([extg, call, evac]) = -20$ |
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| ■ $goodness([call, extg]) = -20$ | ■ $goodness([extg, evac, call]) = -10$ |
| ■ $goodness([evac, extg]) = 0$ | |

Therefore, with the goodness of initial state set to 0, we have:

$$\mathcal{EP}_{util}(\Pi, s_0) = \{[evac], [evac, call], [evac, call, extg], [evac, extg], [evac, extg, call]\}$$

$$\mathcal{EP}_{hedon}(\Pi, s_0) = \{\}$$

Note: Hedonism returns empty set because no plan has a goodness strictly greater than the initial state, which has a goodness of 0.

FORMALISATION OF PROPERTIES

Overview

We want to define properties that can be used to **evaluate and compare** ethical principles.

Based on which properties are verified, we can make an **educated choice** on what principle to follow according to the context/situation.

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The authors define 8 properties which are directly derived from the principles we just saw.

7 more are presented, and are derived from properties found in social choice theory (we will define what this means later).

Consequential Equivalence

We define this as moral actions that produce "good" consequences. It also states that two plans with the same consequences should be considered equally ethical.

If $\pi \in \mathcal{EP}(\Pi, s_0)$ and $state(do(\pi, s_0)) = state(do(\pi', s_0))$, then $\pi' \in \mathcal{EP}(\Pi, s_0)$, for all $\pi' \in \Pi$

Example:

Suppose $[evac, extg]$ and $[extg, evac]$ both result in the same final state (employees safe, fire extinguished). If $[evac, extg] \in \mathcal{EP}(\Pi, s_0)$, then $[extg, evac] \in \mathcal{EP}(\Pi, s_0)$.

Maximal Welfare

An ethically permissible plan must produce at least as much goodness as any other plan that is ethically admissible.

if $\pi \in \mathcal{EP}(\Pi, s_0)$, then for all $\pi' \in \Pi$, $goodness(\pi) \geq goodness(\pi')$

Example:

If $[evac, extg] \in \mathcal{EP}(\Pi, s_0)$ and $[call]$ is a plan (not yet determined if ethical or not), then for $[call]$ to be ethically permissible, it must have at least as much goodness as $[evac, extg]$.

Aggregationism

If a plan is ethically permissible and another plan is even better (i.e. greater goodness), then the better plan must also be ethically permissible.

if $\pi \in \mathcal{EP}(\Pi, s_0)$ and $goodness(\pi') \geq goodness(\pi)$,
then $\pi' \in \mathcal{EP}(\Pi, s_0)$ and for all $\pi' \in \Pi$,

Independance

The permissibility of a plan should not depend on irrelevant alternatives. If a plan is permissible in a larger set, it should remain permissible in any subset containing it.

$$\mathcal{EP}(\Pi, s_0) \cap \Pi' \subseteq \mathcal{EP}(\Pi', s_0)$$

Intuition:

If a plan Π is ethically permissible, and we consider the intersection between this plan and another set of plans Π' , then this intersection should be a subset of the ethically permissible plans in Π' .

Ethical Properties

Beneficence

A permissible plan must be better than doing nothing. In other words, the overall goodness at the end of the plan must be better than the overall goodness at the beginning of the plan.

If $\pi \in \mathcal{EP}(\Pi, s_0)$, then $goodness(\pi) > goodness(\emptyset)$.

Note: this property and Independence are closely tied to Hedonism.

Ethical Properties

New Notations

We define the following terms for the next three properties:

- $harms(\pi) = \{\varphi \mid state(\pi) \vdash \varphi \wedge harmful(\varphi)\}$
- $newState(\pi) = \{F(\vec{x}) \mid F(\vec{x}, do(\pi, s_0)) \wedge \neg F(\vec{x}, s_0)\}$
- $oldState(\pi) = \{F(\vec{x}) \mid F(\vec{x}, do(\pi, s_0)) \wedge F(\vec{x}, s_0)\}$

Positive Responsibility

This property defines that ethical permissibility should be determined only by the new state such actions produce. Together with Harmlessness, this property is closely aligned with the No New Harm principle.

If $\pi \in EP(\Pi, s_0)$ and $newState(\pi) = newState(\pi')$,
then $\pi' \in EP(\Pi, s_0)$, for all $\pi' \in \Pi$.

Negative Responsibility

The property is requiring that an ethically plan should eliminate existing harm or maintain the current good, and is closely aligned with the Remove Harm principle. I.e we want the agent to be responsible for inaction or maintaining harm.

If $\pi \in \mathcal{EP}(\Pi, s_0)$ and $oldState(\pi) = oldState(\pi')$,
then $\pi' \in \mathcal{EP}(\Pi, s_0)$, for all $\pi' \in \Pi$.

Ethical Properties

Harmlessness

This property focuses on the harm of a plan. We should only allow plans that aim to prevent harm, and so the harm at the end of the plan determines its ethical permissibility.

If $\pi \in \mathcal{EP}(\Pi, s_0)$ and $harms(\pi') \subseteq harms(\pi)$,
then $\pi' \in \mathcal{EP}(\Pi, s_0)$, for all $\pi' \in \Pi$.

Social Choice Properties

Overview

Social choice theory designs protocols to select the best outcomes from preferences or choices.

- Aggregating individual preferences into collective decisions,
- Analyzing voting systems,
- Investigating choice functions.

The paper presents seven social choice properties to ensure $\mathcal{EP}$ is robust and consistent.

Consistency Preservation (cp)

This property ensures that if there are any admissible plans, at least one must be ethically permissible. It guarantees that a decision is always made, even in dilemmas.

If $\Pi \neq \emptyset$, then $\mathcal{EP}(\Pi, s_0) \neq \emptyset$.

Social Choice Properties

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Iteration (it)

This property requires that applying the ethical permissibility function twice (or more) does not change the result. It ensures stability in decisions.

$$\mathcal{EP}(\mathcal{EP}(\Pi, s_0), s_0) = \mathcal{EP}(\Pi, s_0).$$

Social Choice Properties

Cut (c)

This property states that if you remove some impermissible plans from Π , the permissibility of the remaining plans does not change. It ensures robustness when the set of plans is reduced.

If $\mathcal{EP}(\Pi, s_0) \subseteq \Pi' \subseteq \Pi$, then $\mathcal{EP}(\Pi, s_0) \subseteq \mathcal{EP}(\Pi', s_0)$.

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Aizerman (aiz)

This property ensures that if you restrict the set of plans to a subset Π' that includes all permissible plans, the result does not change. It guarantees that adding back previously rejected plans does not affect the permissibility of the kept plans.

If $\mathcal{EP}(\Pi, s_0) \subseteq \Pi' \subseteq \Pi$, then $\mathcal{EP}(\Pi', s_0) \subseteq \mathcal{EP}(\Pi, s_0)$.

Note: these two are tightly related, and refer to contextual blindness.

Social Choice Properties

Chernoff (ch)

This property requires that if you shrink the set of plans ($\Pi' \subseteq \Pi$), the permissible plans in Π' must include those that were permissible in Π . It ensures fairness when the set of plans is reduced.

If $\Pi' \subseteq \Pi$, then $\mathcal{EP}(\Pi, s_0) \cap \Pi' \subseteq \mathcal{EP}(\Pi', s_0)$.

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Sen (s)

This property states that if a plan π is permissible in both Π and Π' , then it must be permissible in their intersection $\Pi \cap \Pi'$. It ensures consistency across different contexts.

If $\mathcal{EP}(\Pi, s_0) \cap \mathcal{EP}(\Pi', s_0) \neq \emptyset$, then
 $\mathcal{EP}(\Pi \cap \Pi', s_0) \subseteq \mathcal{EP}(\Pi, s_0) \cap \mathcal{EP}(\Pi', s_0)$.

Social Choice Properties

Arrow (iia)

This property, also known as Independence of Irrelevant Alternatives, requires that if a subset Π' of plans includes some permissible plans from Π , then the permissible plans in Π' are exactly those. It ensures that the permissibility of a plan depends only on how it compares to other available plans.

If $\Pi' \subseteq \Pi$ and $\mathcal{EP}(\Pi, s_0) \cap \Pi' \neq \emptyset$, then
 $\mathcal{EP}(\Pi, s_0) \cap \Pi' = \mathcal{EP}(\Pi', s_0)$.

RESULTS

Results

	Utilitarianism	Hedonism	No New Harm	Remove Harm	Harm Avoidance
Consequential Equivalence	✓	✓	✓	✓	✓
Maximal Welfare	✓	✗	✗	✗	✗
Aggregationism	✓	✓	✗	✗	✗
Independence	✗	✓	✗	✗	✗
Beneficence	✗	✓	✗	✗	✗
Positive Responsibility	✗	✗	✓	✗	✗
Negative Responsibility	✗	✗	✗	✓	✗
Harmlessness	✗	✗	✓	✓	✓
Consistency Preservation	✓	✗	✗	✗	✗
Iteration	✓	✓	✓	✓	✓
Cut	✓	✓	✓	✓	✓
Aizerman	✓	✓	✗	✗	✗
Chernoff	✓	✓	✓	✓	✓
Sen	✓	✓	✓	✓	✓
Arrow	✓	✓	✓	✓	✓

Table 1: Properties of different ethical principles: ✓ indicates satisfaction and ✗ indicates non-satisfaction

Interpretation

Recap

Utilitarianism and Hedonism are motivated by maximising goodness.

The other three principles are motivated by harm reduction.

Utilitarianism is "stronger" (always selects a plan), while Hedonism is "weaker" (rejects non-improving plans).

Harm Avoidance is the strictest but may be too restrictive in real-world dilemmas.

APPENDIX

Full Example Definition

Fluents

- *OnFire(s)*
- *EmployeesSafe(s)*
- *FireExtinguished(s)*
- *FireDepartmentCalled(s)*
- *BuildingDamaged(s)*
- *EmployeesInjured(s)*

Actions

- *extg*
- *evac*
- *call*

Full Example Definition

Initial Definition

- $OnFire(s_0) = True$
- $EmployeesSafe(s_0) = False$
- $FireExtinguished(s_0) = False$
- $FireDepartmentCalled(s_0) = False$
- $BuildingDamaged(s_0) = False$
- $EmployeesInjured(s_0) = False$

Action Precondition Axioms

- $Poss(extg, s) \equiv OnFire(s)$
- $Poss(evac, s) \equiv \neg EmployeesSafe(s)$
- $Poss(call, s) \equiv \neg FireDepartmentCalled(s)$

Positive Effect Axioms

- $\gamma_{FireExtinguished}^+(extg, s) \rightarrow FireExtinguished(do(extg, s))$
- $\gamma_{EmployeesSafe}^+(evac, s) \rightarrow EmployeesSafe(do(evac, s))$
- $\gamma_{FireDepartmentCalled}^+(call, s) \rightarrow FireDepartmentCalled(do(call, s))$

Negative Effect Axioms

- $\gamma_{OnFire}^-(extg, s) \rightarrow \neg OnFire(do(extg, s))$
- $\gamma_{BuildingDamaged}^-(call, s) \rightarrow \neg BuildingDamaged(do(call, s))$
- $\gamma_{EmployeesInjured}^-(evac, s) \rightarrow \neg EmployeesInjured(do(evac, s))$